

Hospital 30-Day Readmission Risk Analysis

A Clinical Data Study

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Executive Summary

This study analyzes 53,122 eligible hospital admissions from the MIMIC-III critical care database to characterize 30-day readmission risk. Overall, 3,384 admissions (6.37%) resulted in a readmission within 30 days of discharge. The strongest signal in the data is payer type: Medicare patients account for 57.74% of all readmissions despite representing only 45.76% of eligible admissions, and the single highest-risk cell is Acute Respiratory Failure patients on Medicare discharged to a skilled nursing facility.

Background

Unplanned 30-day readmissions cost the U.S. healthcare system an estimated \$26 billion annually and are one of the clearest signals of gaps in discharge planning and post-acute care coordination. Under the Hospital Readmissions Reduction Program, the Centers for Medicare & Medicaid Services (CMS) financially penalizes hospitals with excess readmission rates for targeted conditions, making 30-day readmission a core quality metric in value-based care contracts. Reducing avoidable readmissions is therefore both a patient-safety priority and a direct financial one for health systems.

Data & Methodology

The analysis uses the MIMIC-III (Medical Information Mart for Intensive Care III) database, a de-identified clinical dataset covering critical care admissions at Beth Israel Deaconess Medical Center, Boston, released via PhysioNet. Four tables were used: **ADMISSIONS** (admission/discharge timestamps, insurance, discharge disposition), **PATIENTS** (date of birth, used to derive age at admission), **DIAGNOSES_ICD** (primary and secondary ICD-9 diagnosis codes per admission), and **DRGCODES** (diagnosis-related groups).

A readmission was defined using a SQL **LEAD()** window function: admissions were partitioned by patient and ordered chronologically, and each admission was paired with only its immediate next admission. An admission counted as a 30-day readmission if that next admission began after discharge and within 30 days of it, and if the index admission did not end in an in-hospital death. An earlier version of this analysis used a self-join of **ADMISSIONS** against itself on patient ID; for patients with three or more admissions, that approach matched every later admission against every earlier one rather than only the next one, double-counting readmissions and inflating downstream breakdowns. The **LEAD()**-based approach was adopted specifically to eliminate that double counting.

Key Findings

- **Finding 1 — Overall readmission rate.** 6.37% overall 30-day readmission rate: 3,384 readmissions out of 53,122 eligible admissions.
- **Finding 2 — Medicare is overrepresented.** Medicare patients account for 57.74% of all readmissions, compared to 45.76% of eligible admissions overall — a disproportionate share relative to their size in the population.
- **Finding 3 — Home-based discharges dominate.** HOME HEALTH CARE and HOME discharges together account for 47.49% of readmissions, indicating that patients sent home (with or without home health support) are readmitted most often.
- **Finding 4 — Highest-risk combination.** Acute Respiratory Failure + Medicare + discharge to a Skilled Nursing Facility (SNF) is the single largest risk cell in the data: 30 cases, averaging 11.0 days to readmission.
- **Finding 5 — Age group breakdown.** Readmissions are fairly evenly spread across age groups, with a mild concentration in late-middle-age and older patients: 66–80 years is the largest group at 28.16% (953 cases), followed by 50–65 at 25.86% (875), Under 50 at 25.24% (854), and Over 80 at 20.74% (702).
- **Finding 6 — Predictive model performance.** A logistic regression model trained on insurance type and discharge location alone (one-hot encoded, class-balanced) achieved 53.66% accuracy predicting 30-day readmission. This modest accuracy indicates these two administrative variables carry real but limited predictive signal on their own; incorporating clinical features (vitals, labs, prior admission count, length of stay) would likely improve performance.

Clinical Recommendations

- **Recommendation 1 — Targeted medication reconciliation.** Pharmacist-led medication reconciliation call at day 7 post-discharge for Medicare patients discharged home with cardiac or respiratory diagnoses, the population and time window where risk is most concentrated.
- **Recommendation 2 — SNF follow-up for respiratory patients.** Weekly pharmacist check-ins for Acute Respiratory Failure patients discharged to a skilled nursing facility, the single highest-risk combination identified in this analysis.
- **Recommendation 3 — Simplified regimens for Medicaid CKD patients.** Simplified medication regimens and community pharmacy partnerships for Medicaid patients with chronic kidney disease, to reduce regimen complexity as a barrier to adherence after discharge.

Limitations & Future Work

- MIMIC-III is ICU data from a single Boston hospital — findings may not generalize nationally or to non-ICU populations.
- The readmission definition captures any return visit within 30 days, not only unplanned readmissions, which may include some planned follow-up procedures.

- Future work: ICD-10 diagnosis groupings, length-of-stay-by-age analysis, and an expanded predictive model incorporating clinical features (vitals, labs, prior utilization) and external validation on a second dataset.

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